COM5231 · Introduction to Wireless and Ad Hoc Networking

Part 3 — Study Notes

Wireless LANs · Bluetooth/802.15/ZigBee · Routing and MANETs

SUPPLEMENTARY content not found in the uploaded lecture notes

NOTES ERROR a mistake in the lecture notes, with the verified correction

EXAM how the topic is or could be examined

★ high-priority, frequently examined content

Exam preparation pack, built from the uploaded lecture materials and past papers

Part 3 — Detailed Study Notes: Topics 9–11

Topics in this part 9. Wireless LANs: IEEE 802.11 10. Bluetooth, IEEE 802.15 and ZigBee 11. Routing and Ad Hoc Network Routing (SUPPLEMENTARY: no lecture notes were provided)

Source codes used in this part

Code	File
[WL n]	*WLANS_v2.pptx*, slide n
[FZ]	*Chapter 15 WLANS by Forouzan.pdf*
[BT-B]	*BLUETOOTH AND IEEE 802.15_V2.pdf* (Beard & Stallings Ch. 12)
[BT-S n]	*Bluetooth and IEEE 802.15.pptx*, slide n (image-only slides, read visually)
[BTX n]	*Bluetooth_Exercises_Reordered_20_Questions.pdf*, question n
[AT]	*Fundamentals of Communications Access Technologies.pdf*
[WCP]	*Wireless Communication Problems.pdf*

TOPIC 9 — WIRELESS LANs (IEEE 802.11)

9.1 Introduction

A **LAN** interconnects computers within a small area (home, school, lab, office building) using network media. A **WLAN** links two or more devices using a **wireless medium** within such an area. It is sometimes called **wireless Ethernet**, and in the USA and elsewhere the public uses **WiFi** (*wireless fidelity*) as a synonym [WL 1]. Strictly, WiFi is a WLAN **certified by the WiFi Alliance**, a non-profit association of 300+ companies [FZ].

WLANs are a fast-growing market that brings the flexibility of wireless access into office, home and production environments. They are typically restricted to buildings, a campus or single rooms. **Goal: replace office cabling, and add flexibility for ad hoc communication** (e.g. group meetings) [WL 2].

Exam weight: very high. Hidden node, CSMA/CA, ACKs, two ISPs on channel 11, wired vs wireless LANs and co-channel interference together account for about **20 marks of the exam** and much of Test 1 and Test 2.

9.2 Key definitions

Term	Definition	Source
BSS (Basic Service Set)	The building block of a WLAN: stationary or mobile wireless stations and an **optional** central base station, the **access point (AP)**	[WL 10], [FZ]
Ad hoc BSS	A BSS **without an AP**: a stand-alone network that **cannot send data to other BSSs**; stations send frames directly to each other	[WL 10], [FZ]
Infrastructure BSS	A BSS **with an AP**; each client associates with an AP that connects to other networks	[WL 11]
ESS (Extended Service Set)	Two or more BSSs with APs, connected through a **distribution system** (wired or wireless, usually Ethernet)	[WL 11], [FZ]
Access point (AP)	A radio that is a **wireless portal**: it **controls medium access** and acts as a **bridge** to other wireless or wired networks	[WL 5], [FZ]
SSID	**Service set identifier**: the logical **name** of an 802.11 network	[FZ]
DCF	**Distributed coordination function**: the fundamental 802.11 MAC mechanism, based on **CSMA/CA**	[WL 19]
PCF	**Point coordination function**: **optional, centralized, contention-free** polling by the AP (infrastructure only)	[WL 19, 40]
NAV	**Network allocation vector**: a timer showing how long other stations must wait before checking whether the channel is idle	[WL 37]
IFS	**Interframe space**: a waiting time before transmitting. Its length sets the **priority** (SIFS < PIFS < DIFS)	[WL 34]

9.3 WLAN characteristics, design goals, wired vs wireless

Advantages and disadvantages [WL 3]

Advantages	Disadvantages
Very flexible within radio coverage	**Low bandwidth** compared to wired (~11–54 Mbit/s in the slide), because of radio limits, **higher error rates** from interference, and **higher delay/delay variation** from extensive error correction and detection
Ad hoc networks without previous planning	**Lower QoS**
Allows small, **independent devices**	**Proprietary solutions**; standards take time (slow standardisation)
More robust against disasters (earthquakes, fire)	Enhanced features only work in a **homogeneous** environment (same-vendor adapters)
	Must follow many **national restrictions**; global solutions are slow

Design goals [WL 4]

Global, seamless operation; **low power** for batteries (power-saving modes); **no special permissions or licences** (licence-free band); **robust** transmission; **easy to use**, simple management; **protection of investment in wired networks** (same data types and services); **security** (no one reads others' data), **privacy** (no user profiles collected), **safety** (low radiation).

★ Wired LAN vs wireless LAN [FZ] (Asked in: T2-2025 Q5, 10 marks)

Aspect	Wired LAN	Wireless LAN
Medium	Wires; dedicated/switched links	**Air**; the signal is **broadcast**, all hosts **share** the medium (multiple access)
Hosts	Connected at a fixed point, with a link-layer address tied to the NIC; must be physically connected	**Not physically connected**; can **move freely** while using the network
Isolated LAN	Hosts connected via a **link-layer switch**	**Ad hoc network**: hosts communicate freely, **no AP**
Connection to other networks	Via a **router**	Via an **access point** that **glues two environments** (wireless ↔ wired); different role from a switch
Moving between them	Only the lower two layers differ. Change the NICs and replace the switch with an AP: **MAC addresses change, IP addresses stay the same**	
Access control	**CSMA/CD**	**CSMA/CA** (CD doesn't work in wireless)
Characteristics	Negligible attenuation and interference	**Attenuation, interference, multipath propagation, higher error rate**

★ Unique concerns a wireless LAN designer must address (T2-2025 Q5, second half)

From the notes [FZ characteristics; WL 3–4; WL 16, 29; WCP]:

1. **Attenuation / path loss**: signal strength drops quickly with distance and through walls → place APs carefully, add APs for coverage [WCP path loss].
2. **Interference**: co-channel and adjacent-channel interference from other WLANs, Bluetooth and microwave ovens in the crowded **2.4 GHz ISM band** → channel planning, 5 GHz [WL 23, WCP].
3. **Multipath propagation and fading** → equalisation, antenna diversity, MIMO, OFDM, lower data rates [WCP].
4. **Higher error rate** → ACKs and retransmission, FEC, **fragmentation** [FZ].
5. **Medium access without collision detection**: hidden and exposed stations → CSMA/CA, RTS/CTS, NAV [FZ].
6. **Mobility**: hosts move between BSSs/ESSs (BSS-transition, ESS-transition mobility) → association, handoff [FZ].
7. **Limited power / battery** → power-saving modes [WL 4].
8. **Security and privacy**: broadcast medium, anyone in range can listen → authentication and encryption [WL 4, 27].
9. **Licence-free spectrum regulation**: power limits (≤ 1 W, typically 50 mW), national restrictions [WL 23, WL 3].
10. **QoS and capacity**: shared, lower bandwidth → rate adaptation, PCF for time-sensitive traffic [WL 6, 23, 40].

9.4 Infrastructure vs ad hoc networks [WL 5–9]

Infrastructure-based:

- Provides **access to other networks**. Communication is typically **only between wireless nodes and the AP**, not directly between nodes.
- The AP **controls medium access** and acts as a **bridge**.
- Several BSSs + a fixed network can form **one logical network** beyond radio coverage.
- Network functionality lies in the AP, so **clients stay simple**.
- If **only the AP controls access, no collisions** are possible, which is useful for **QoS** (the AP can **poll** nodes to guarantee data rates).
- **Loses flexibility**: can't be used for disaster relief if no infrastructure is left.

Ad hoc:

- **No infrastructure**, so the **greatest flexibility**; no AP controls access.
- **Each node is more complex**: it implements medium access and **forwards data**.
- Nodes communicate **only if within each other's radio range, or if other nodes forward the message**.
- **802.11 and HiperLAN2** are typically infrastructure-based but also support ad hoc. **Bluetooth is a typical ad hoc network.** (HiperLAN = European high-performance radio LAN standard, similar to WiFi.)

9.5 IEEE 802.11 architecture [WL 10–14, FZ]

- **BSS** (ad hoc or infrastructure) and **ESS** (BSSs connected by a **distribution system**). The ESS has **mobile stations** (normal stations inside a BSS) and **stationary stations** (APs that are part of the wired LAN).
- Stations within reach can communicate **without the AP**. Communication **outside the BSS goes via the AP**. This is like a cellular network: **BSS ≈ cell, AP ≈ base station** [WL 11 notes].

- A mobile station **can belong to more than one BSS** at the same time [FZ].

Station types by mobility [FZ]:

Type	Meaning
No-transition	Stationary, or moving only inside one BSS
BSS-transition	Moves between BSSs **within one ESS**
ESS-transition	Moves from one ESS to another; **802.11 does not guarantee continuous communication** during the move

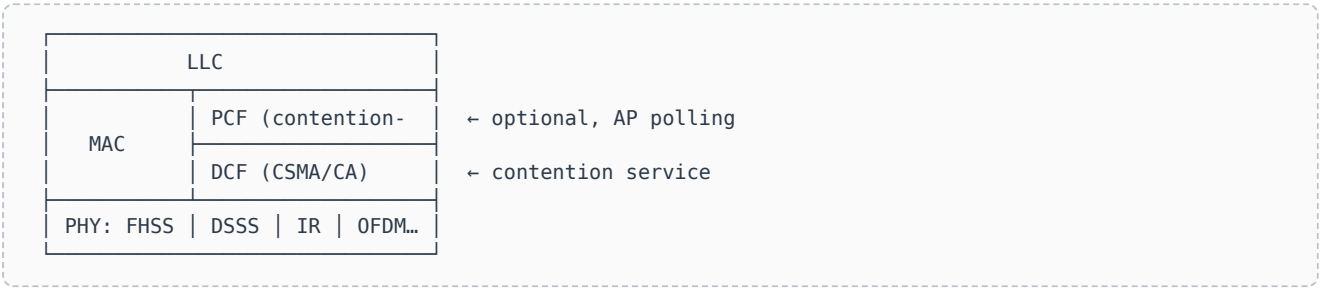
Channels and association [WL 12–13]

- The spectrum is **divided into channels** at different frequencies. The **AP administrator chooses the AP's channel**. **Interference is possible: a neighbouring AP may choose the same channel!**
- An arriving host must **associate** with an AP:
  - it **scans channels** listening for **beacon frames** containing the AP's **SSID and MAC address**;
  - it **selects** an AP and may perform **authentication**.
- **Passive scanning**: (1) APs send beacon frames; (2) the host sends an **association request** to the selected AP; (3) the AP sends an **association response**.
- SUPPLEMENTARY: **active scanning**: the host broadcasts a **probe request**, APs reply with **probe responses**, then association request/response.

SUPPLEMENTARY (needed for the channel 1/11 question): the 2.4 GHz band (2.4–2.4835 GHz) has **11 channels** (US) spaced **5 MHz** apart, but each 802.11b/g signal is about **22 MHz wide** (see [WC-S 4]: a 22 MHz signal at 2.412 GHz occupies 2.401–2.423 GHz). So adjacent channels **overlap**, and only **channels 1, 6 and 11 are non-overlapping**.

9.6 Protocol architecture [WL 15–19, FZ]

- 802.11 covers the **physical and data-link layers**.
- The data-link layer is split into:
  - **LLC (logical link control)**: framing, flow and error control; a **common interface** between the network layer and the 802.11 MAC; identifies/multiplexes higher-layer protocols and **hides MAC details**.
  - **MAC (medium access control)**: decides **who transmits next**. 802.11 defines two MAC sublayers: **DCF** (bottom, contention, CSMA/CA) and **PCF** (top, optional, contention-free polling).
- **Three classes of multiple access protocols**: **channel partitioning** (incl. CDMA), **random access**, **taking turns**. 802.11's random access protocol is **CSMA/CA** [WL 15].



9.7 Physical layer [WL 21–26]

- Bands: **2.4–2.485 GHz** and **5.1–5.8 GHz**. The 2.4 GHz band is **unlicensed** and shared with phones and **microwave ovens**. At 5 GHz, range is **shorter** for a given power and **multipath** is worse, but there is **less interference**.
- All except infrared use **short-range radios in the ISM bands**: unlicensed, with a radiated power of **at most 1 W (50 mW typical)**. Garage door openers, cordless phones and microwaves compete for the same spectrum. **2.4 GHz is more crowded than 5 GHz.**
- **Rate adaptation**: each PHY defines **multiple rates**. A weak signal uses a **low rate**; a clear signal uses the **highest** rate.
- 802.11n, ac and ax use **MIMO**.

Standard	Technique	Band	Modulation	Data rate
802.11 FHSS	Frequency hopping (79 × 1 MHz sub-bands)	2.4 GHz ISM	2- or 4-level FSK	1 or 2 Mbps
802.11 DSSS	Direct sequence	2.4 GHz ISM	BPSK or QPSK	1 or 2 Mbps
802.11 Infrared	Infrared light, 800–950 nm (not ISM)	—	**Pulse position modulation (PPM)**	1 or 2 Mbps
802.11a	**OFDM** (with QAM)	**5 GHz** (5.725–5.850 GHz)	QAM	up to **54 Mbps**
802.11b	**HR-DSSS**	2.4 GHz	—	**5.5 and 11 Mbps**
802.11g	**OFDM + FEC**	2.4 GHz (compatible with b)	—	**22 or 54 Mbps**
802.11n	**MIMO** (+ OFDM)	2.4/5 GHz	—	up to **600 Mbps**

> **NOTES ERROR**: [WL 24–26] write the 2.4 GHz band as "2.400–**4.835** GHz". It should be **2.400–2.4835 GHz** (the same slides correctly say 2.4–2.485 GHz on slide 21).

9.8 MAC functions [WL 27–28]

The 802.11 MAC covers three areas: **reliable data delivery, access control, security**.

Reliable data delivery: the physical and MAC layers are **unreliable** (noise, interference, propagation effects), so many frames are lost. Reliability could be left to higher layers such as **TCP**, but 802.11 adds a **frame exchange protocol at the MAC layer**:

- **Two-frame exchange**: the receiver returns an **ACK**. If the source gets no ACK within a short time (the data or the ACK was damaged), it **retransmits**.
- **Four-frame exchange** (more reliable): **RTS → CTS → DATA → ACK**.
  - **RTS** alerts all stations **within range of the source** that an exchange is under way, so they refrain from transmitting.
  - **CTS** alerts all stations **within range of the destination**.

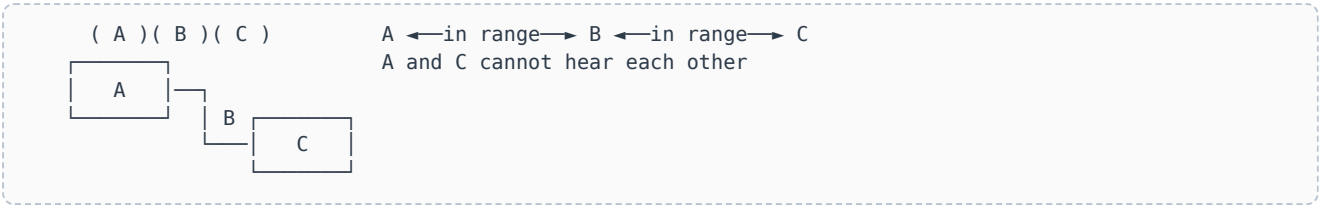
9.9 ★ Why not CSMA/CD in wireless? [WL 16, WL 29, FZ]

Wireless LANs **cannot implement CSMA/CD** for three reasons:

1. **Send and receive at the same time**: to detect a collision a station must transmit and listen simultaneously (full duplex). That means **costly stations and more bandwidth**. Battery-powered hosts don't have the power. Also, **received signals are much weaker than transmitted ones** [WL 16].
2. **Hidden station problem**: collisions may happen **without being detected**.
3. **Distance / fading**: stations can be far apart; **signal fading** may prevent a station at one end from hearing a collision at the other end.

9.10 ★ The hidden station (hidden terminal) problem [FZ, WL 30, 39, 44]

Using the exam's **Figure 2**: three nodes **A, B, C** in a row. Each circle is a node's transmission range. **B is within range of both A and C, but A and C are out of each other's range.**



- **Problem**: A senses the channel, hears nothing (C is out of range), and starts sending to B. C also senses an idle channel (it can't hear A) and starts sending to B. **Both frames collide at B**, but **neither A nor C can detect it**. A and C are **hidden from each other with respect to B**.
- **Consequences**: frames are lost, retransmissions follow, and **capacity is reduced** [FZ]. Hidden stations can also be caused by **obstacles** [FZ].
- **What happens if A and C send simultaneously** (Exam Q5(b), T1A Q8): both start transmitting because each senses the medium idle. Their signals **overlap at B**, so B receives a **garbled** frame and decodes neither. A and C keep transmitting their whole frames (no collision detection), wasting the channel. Neither gets an **ACK**, so both time out, **back off a random time and retransmit**.

★ Solution: RTS/CTS handshake [FZ, WL 39, 43]

- Before sending data, the source sends an **RTS** that includes the **duration** it needs. The destination replies with a **CTS** that also carries the duration.
- **Key point**: the **CTS** is heard by **everyone in range of the receiver**, including the hidden station. In Forouzan's figure (B sending to A, with C hidden from B): B's RTS reaches A but not C. Because both B and C are in range of A, **A's CTS reaches C**. C learns that a hidden station is using the channel and **refrains until that duration is over**.
- *"The CTS frame in the CSMA/CA handshake can prevent collision from a hidden station"* [WL 43].

9.11 ★ CSMA/CA and DCF: the complete procedure [WL 32–35, FZ]

Interframe spaces [WL 34]. A shorter IFS means **higher priority**:

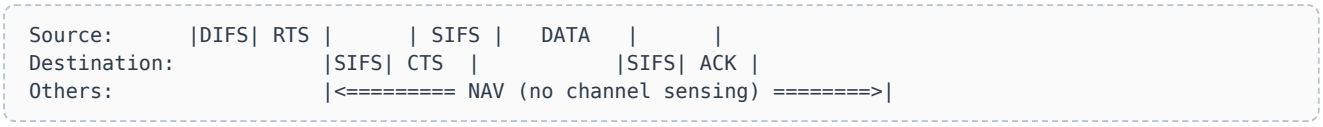
IFS	Length	Used for
SIFS (short IFS)	**Shortest** (highest priority)	**Immediate responses**: ACK after data, CTS after RTS, data after CTS

IFS	Length	Used for
PIFS (PCF IFS)	Middle	The **point coordinator (AP)** issuing **polls** in PCF; takes precedence over contention traffic
DIFS (DCF IFS)	**Longest**	**Ordinary (contention) traffic**

DCF algorithm [WL 32–33, 35]:

1. A station with a frame **senses the medium** (checks the energy level at the carrier frequency [FZ]).
2. If **idle**, it waits **one IFS (DIFS)**. If still idle, it may transmit.
3. If **busy** (initially or during the IFS), it **defers** and monitors until the current transmission ends.
4. After that, it waits **another DIFS**. If idle, it waits a **random backoff** time and senses again.
5. **During backoff**, if the medium becomes busy, the **backoff timer freezes** and **resumes** when the medium becomes idle. When the counter reaches **zero**, the station transmits.
6. Optional **RTS → (SIFS) → CTS → (SIFS) → DATA → (SIFS) → ACK** exchange.
7. Other stations set their **NAV** and defer.
8. **No ACK (or no CTS) = assume a collision**: perform another backoff and retry.

Frame exchange timeline [FZ Fig 15.7]



Network allocation vector (NAV) ★ [WL 37, FZ]

How is collision avoidance accomplished? The key is NAV. The RTS includes the **duration** the sender needs. Stations that hear it set a **NAV timer** showing how long to wait before even **checking the channel**. Before sensing the physical medium, each station **first checks whether its NAV has expired**. This is **virtual carrier sensing**.

Collision during handshaking [WL 38, FZ]

Two stations may send **RTS frames at the same time**, and they collide. There is **no collision detection**, so a sender **assumes a collision if no CTS arrives**, applies the **backoff strategy**, and tries again. (RTS frames are short, so a collision wastes little time.)

★ Applying CSMA/CA to Figure 2: A sends a frame to B (Exam Q5(c), T1A Q9)

1. A **senses** the medium (and its NAV). It is idle, so A waits **DIFS**, then (if still idle) completes a **random backoff**.
2. A sends an **RTS** to B with the duration of the whole exchange. **B** hears it (C does not).
3. B waits **SIFS** and replies with a **CTS** that also carries the duration. **Both A and C hear the CTS** (C is in B's range). **C sets its NAV** and stays silent, even though it can't hear A.
4. A waits **SIFS** and sends **DATA** to B.
5. B waits **SIFS** and returns an **ACK**. When the NAV expires, the medium is free and C may contend (DIFS + backoff).
6. If A gets **no CTS or no ACK**, it assumes a collision, **doubles its backoff window** (SUPPLEMENTARY: binary exponential backoff) and retries.

9.12 The exposed station problem [FZ, WL 31, 45–46]

- **A is transmitting to B. C wants to send to D**, which would **not** interfere with A→B. But C is **within A's range**, so it hears A's transmission and **refrains**. **C is too conservative and wastes capacity.**
- **RTS/CTS does not help**: C hears A's RTS and stays quiet, because it cannot know that its transmission wouldn't affect the zone between A and B.

```
D ⟵ C        A ⟶ B
      (C hears A, so stays silent, although C→D would not hurt B)
```

9.13 Point coordination function (PCF) [WL 40–42, FZ]

- **Optional**; only in **infrastructure** networks (not ad hoc); implemented **on top of DCF**; used mostly for **time-sensitive** traffic.
- **Centralised polling**: the **AP (primary)** polls stations (**secondaries**) one after another, and they send any data they have **to the AP**. All exchanges go through the primary, which decides who uses the channel. **Poll and select** functions prevent collisions.
- **Drawback**: if the **primary station fails, the system goes down**.
- **Priority over DCF**: PCF uses **PIFS**, which is **shorter than DIFS**.
- **Repetition interval** [WL 42; SUPPLEMENTARY detail]: time is divided into repeating **superframes**. Each starts with a **beacon**, followed by a **contention-free period** (PCF polling) and then a **contention period** (DCF), so both kinds of traffic coexist.

9.14 Fragmentation [FZ]

The wireless environment is **very noisy**, so frames are often corrupted. 802.11 recommends **fragmentation**: dividing a large frame into smaller ones, because **resending a small frame is more efficient than resending a large one**.

9.15 ★ Why acknowledgements in 802.11 but not wired Ethernet? (Exam Q5(d), T1A Q10)

- In **CSMA/CD Ethernet**, collision detection serves two purposes: a **detected** collision means resend, and the **absence** of a collision acts as **a kind of acknowledgement** that the frame got through [FZ]. Wired links also have **very low error rates**.
- In **802.11**, stations **cannot detect collisions** (no simultaneous send/receive, hidden stations, fading) [WL 16, 29]. And wireless frames are often lost to **noise, interference and propagation effects** [WL 27]. *"Acknowledgment is needed in this protocol because the station does not have any means to check for the successful arrival of its data at the destination"* [FZ]. So the receiver **returns an ACK**, and **no ACK = retransmit** [WL 27].

9.16 ★ Co-channel interference in WiFi (T1A Q13, 4 marks)

Why many WiFi networks have a **high error rate** from co-channel interference:

- The AP administrator **chooses the channel**, and **neighbouring APs may choose the same channel** [WL 12].

- **2.4 GHz is crowded**: few non-overlapping channels (1, 6, 11 — SUPPLEMENTARY), shared with cordless phones, Bluetooth and **microwave ovens** [WL 21, 23].
- **Inward / co-channel interference** comes from products using **similar radio frequencies** nearby, which **block transmissions or cause bit errors** [WCP].
- **ISM users have no regulatory protection** and must tolerate interference [WC-S 10].
- Hidden stations in other BSSs on the same channel cause **collisions that CSMA/CA can't prevent**.
- **Result**: corrupted frames, failed ACKs, retransmissions, backoffs and lower throughput.

9.17 Topic 9 revision summary

- WLAN = wireless LAN = "wireless Ethernet" = WiFi (certified). **BSS** (ad hoc or infrastructure with AP), **ESS** = BSSs + distribution system; SSID; station mobility types.
- Association: scan for **beacons** (SSID, MAC) → association request/response. AP channel is chosen by the admin, and neighbours may clash.
- Stack: **LLC + MAC (DCF/CSMA/CA + optional PCF)**; PHY variants (FHSS, DSSS, IR/PPM, a/g OFDM 54 Mbps, b HR-DSSS 11 Mbps, n MIMO 600 Mbps). 2.4 GHz crowded; 5 GHz shorter range, less interference. **Rate adaptation.**
- **No CSMA/CD**: can't send+receive; hidden station; fading.
- **CSMA/CA**: sense → DIFS → random backoff (frozen while busy) → RTS/SIFS/CTS/SIFS/DATA/SIFS/ACK. **NAV** = virtual carrier sense. **SIFS < PIFS < DIFS.** No CTS/ACK → backoff and retry.
- **Hidden station**: A and C both reach B but not each other → collision at B. **RTS/CTS fixes it**: the CTS reaches the hidden node.
- **Exposed station**: C needlessly defers; RTS/CTS can't fix it.
- **PCF**: AP polling, PIFS, infrastructure only, single point of failure.
- **ACKs**: no collision detection plus high error rates.

TOPIC 10 — BLUETOOTH, IEEE 802.15 AND ZIGBEE

10.1 Introduction: short-range communication and the IoT [BT-B 12.1, BT-S 2–4]

Short-range technologies (a few metres) focus on **low cost and low energy** for long battery life. Several standards come from the **IEEE 802.15 working group**. They provide two services:

1. **Personal area network (PAN)**: devices within a person's working space, especially **on their bodies**, share information and communicate with the user.
2. **Internet of Things (IoT)**: **device-to-device** communication for **data collection, analysis and automation**. Examples: heart-monitoring implants, biochip transponders on farm animals, cars with sensors, devices that assist fire-fighters [BT-S 3]. IoT flow: **sensor → connectivity → data processing → user interface**, and the user can also act back on the system (e.g. adjust cold-storage temperature from a phone) [BT-S 4].

10.2 Bluetooth overview [BT-B 12.2, BT-S 6–8]

- An **always-on, short-range radio hookup on a microchip**. Developed by **Ericsson in 1994** so laptops could make calls via a mobile phone. Originally published as **IEEE 802.15.1**; now the **Bluetooth SIG** (special interest group) publishes the standards. Named after King **Harald Blaatand** of Denmark.
- **Key attributes: robustness, low power consumption, low cost.**
- **2.4 GHz** (globally available, unlicensed, low power). Two devices **within 10 m** can share **up to 2.1 Mbps or 24 Mbps**.
- Supports an open-ended list of applications: data, audio, graphics, video.
- **Up to 8 devices in a piconet**; **10 piconets can coexist** in the same coverage range; each link is **encoded** against eavesdropping and interference.

Three application areas [BT-B, BT-S 8]:

1. **Data and voice access points**: real-time voice and data, wirelessly connecting portable and stationary devices.
2. **Cable replacement**: removes proprietary cables. Connections are instant and work **without line of sight**. Range **≈ 10 m**, extendable to **100 m** with an amplifier.
3. **Ad hoc networking**: a device can connect instantly to another Bluetooth radio as soon as it comes into range.

Uses (Table 12.1): hands-free headset, car stereo, PC input/output, synchronisation, cable replacement (RS-232), infrared remote replacement, low-rate data transfer, bridges (Ethernet interconnection), game consoles, medical devices, security (theft prevention when the link breaks).

★ Why Bluetooth is a WPAN rather than a WLAN (BTX 1)

Its **range is about 10 m** (a person's personal space), it is designed for **low power and low cost** and **cable replacement** between a person's own devices, and its basic network (a piconet) holds only **8 active devices**. A WLAN covers a building or campus (tens to hundreds of metres), serves many users, and connects to wider networks.

10.3 Standards documents: core and profiles [BT-B]

The standards (2500+ pages) have two groups:

- **Core specifications**: details of the protocol layers, from the radio interface to link control, interoperability, testing and timers.
- **Profile specifications**: the use of Bluetooth to support **particular applications**. Each profile says which parts of the core are **mandatory, optional or not applicable** for a usage model. **Purpose: a standard of interoperability**, so products from different vendors that claim a usage model **work together**.

10.4 ★ Protocol architecture [BT-B, BT-S 10–20]



Core protocols (five-layer stack):

Layer	Role
Radio	Air interface: **frequency, frequency hopping, modulation, transmit power**
Baseband	**Connection establishment within a piconet, addressing, packet format, timing, power control**
LMP (link manager protocol)	**Link setup** and ongoing link management, incl. **security (authentication, encryption)** and negotiation of baseband packet sizes
L2CAP	**Adapts upper-layer protocols to the baseband**; provides **connectionless and connection-oriented** services
SDP (service discovery)	Device information, services and their characteristics can be **queried** so a connection can be set up

Cable replacement protocol: RFCOMM presents a **virtual serial port**, making replacement of serial cables as transparent as possible. It provides binary data transport and **emulates EIA-232 (RS-232) control signals** over the baseband. **Telephony control protocol: TCS BIN** (telephony control specification, binary) is a **bit-oriented** protocol for **call control signalling** for speech and data calls, plus **mobility management** for groups of TCS devices. **Adopted protocols** (Bluetooth invents only what's necessary and reuses existing standards):

- **PPP**: carries IP datagrams over a point-to-point link.
- **TCP/UDP/IP**: the foundation protocols of TCP/IP.
- **OBEX**: object exchange; a session-level protocol from **IrDA**, similar to HTTP but simpler. Content formats include **vCard** (business card) and **vCalendar**.

10.5 ★ Profiles [BT-B, BT-S 21]

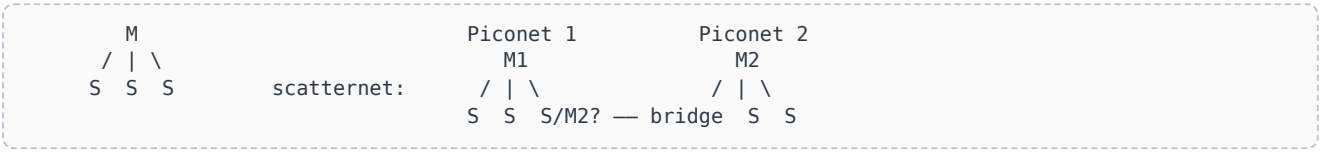
- **Over 40 profiles**. A profile is **a set of protocols that implement a particular Bluetooth-based application**; each defines the protocols and features that support a **usage model**.
- **All Bluetooth nodes support the Generic Access Profile**; profiles may depend on other profiles.
- **Example: the File Transfer Profile** transfers directories, files, documents, images and streaming media, and can browse folders on a remote device. It **depends on the Generic Object File Exchange, Serial Port and Generic Access profiles**, uses **OBEX** within Generic Object File Exchange, and interfaces with **L2CAP and RFCOMM**.

★★ How profiles benefit applications despite adding complexity (Asked in: Exam Q4(c), T2-2025 Q7, T2-2026 Q7, BTX 19)

1. **Interoperability**: the purpose of a profile is a *"standard of interoperability so that products from different vendors that claim to support a given usage model will work together"* [BT-B]. A headset from vendor X works with a phone from vendor Y, so users don't care about brands.
2. **Ready-made building blocks for developers**: the profile already specifies **which protocols and features to use** (mandatory/optional/not applicable). Developers **don't redesign** the protocol stack for each application; they implement the profile, e.g. the File Transfer Profile reuses OBEX, RFCOMM and L2CAP [BT-B].
3. **Only what's needed is implemented**: an application implements only the subset of the large core specification its usage model needs. This means **smaller, cheaper, low-power devices** (key Bluetooth attributes).
4. **Reuse through dependencies**: profiles build on others (all on the Generic Access Profile), so common functions (discovery, connection setup, security) are **consistent** across all applications.
5. **Plug-and-play user experience**: devices discover each other's services (**SDP**) and connect automatically, because both sides know exactly how the application behaves.
6. **Predictable behaviour and testing**: certification against profiles makes products reliable and easier to test.

10.6 ★ Piconets and scatternets [BT-B, BT-S 23]

- **Piconet**: the basic unit, **one master + one to seven active slaves**.
- The **master determines the channel (frequency-hopping sequence) and phase (timing offset, i.e. when to transmit)** for all devices, **using its own device address** as a parameter. Slaves **tune to the same channel and phase**.
- **A slave may only communicate with the master, and only when granted permission by the master.**
- **Scatternet**: a device in one piconet may also be part of **another** piconet, as **slave or master in each**. This overlap is a scatternet.



(A device can be **slave in both**, or **master in one and slave in the other**, but **not master in both**: see 10.12.)

- **Advantage**: many devices share the same physical area and use bandwidth efficiently.

- **Frequency hopping**: **1 MHz carrier spacing**, typically up to **80 frequencies (80 MHz)**. Without hopping, a channel = one 1 MHz band. With hopping, a **logical channel = a hopping sequence**. At any time the available bandwidth is **1 MHz, shared by at most 8 devices**. **Different logical channels share the same 80 MHz. Collisions occur when devices in different piconets happen to use the same hop frequency at the same time**; more piconets means more collisions and poorer performance.
- **Summary**: *the physical area and total bandwidth are shared by the scatternet; the logical channel and data transfer are shared by a piconet.*

Can a piconet have more than eight stations? (BTX 5)

Only **8 can be active** (1 master + 7 active slaves): the active-member address **AM_ADDR is 3 bits** (2^3 = 8 values; 0 = broadcast, so 1–7 for slaves) [BT-S packet header]. SUPPLEMENTARY: additional devices can be **parked** (synchronised but inactive), up to 255, and activated by swapping with an active slave.

10.7 Radio specification [BT-B Table 12.2, BT-S 24–25]

Parameter	Basic Rate (BR)	Enhanced Data Rate (EDR)
Topology	Up to 7 simultaneous links, logical star	Same
Modulation	**GFSK**	**π/4-DQPSK and 8DPSK**
Peak data rate	**1 Mbps**	**2 and 3 Mbps**
RF bandwidth	220 kHz (−3 dB), 1 MHz (−20 dB)	Same
RF band	**2.4 GHz ISM**	Same
RF carriers	**23/79**	23/79
Carrier spacing	**1 MHz**	1 MHz
Transmit power	0.1 W	0.1 W
Piconet access	**FH-TDD-TDMA**	Same
Hop rate	**1600 hops/s**	Same
Scatternet access	**FH-CDMA**	Same

- **Power classes**: **Class 1**: up to **100 mW (+20 dBm)**, min 1 mW; **power control mandatory** (4–20 dBm); greatest range. **Class 2**: up to **2.5 mW (+4 dBm)**, min 0.25 mW (−6 dBm); power control optional. **Class 3**: **max 1 mW**, lowest power.
- **79 × 1 MHz physical channels** in most countries; **f = 2.402 + n MHz, n = 0…78** (US, Europe). Japan, Spain and France have 23 channels.
- **Power control** keeps devices from emitting more than necessary; implemented via **LMP** between master and slaves.
- **BR modulation: Gaussian FSK**. Binary 1 = positive frequency deviation, 0 = negative; **1 Msymbol/s × 1 bit = 1 Mbps**; minimum deviation 115 kHz.
- **EDR** (Bluetooth 2.0): the same 1 Msymbol/s with **π/4-DQPSK (2 bits) and 8DPSK (3 bits)** → 2 and 3 Mbps. The raw 3 Mbps gives **≈ 2.1 Mbps** after overhead.

Answers to BTX 10–13: the radio layer specifies the air interface (frequency, hopping, modulation, power). The actual bandwidth used is **1 MHz per hop channel** within the 79–80 MHz band. The spread-spectrum technique is **FHSS** (1600 hops/s). Modulation is **GFSK** (BR), with π/4-DQPSK/8DPSK for EDR. Bits become radio waves by shifting the carrier frequency up (1) or down (0).

10.8 ★ Baseband: frequency hopping and TDD [BT-B, BT-S 26–30]

FH serves two purposes:

1. **Resistance to interference and multipath** effects.
2. **Multiple access among co-located devices in different piconets.**

FH workflow: total bandwidth is divided into **79 physical channels of 1 MHz** → hopping jumps between them in a **pseudorandom sequence** → **the same sequence is shared by all devices in a piconet** (an "FH channel").

- **1600 hops/s**, so each channel is occupied for **0.625 ms = one slot**; slots are numbered.
- **Time division duplex (TDD)**: data go **one direction at a time**, alternating (master → slave, slave → master). With more than two devices, the access technique is **TDMA**. So piconet access = **FH-TDD-TDMA**.
- **Multislot packets**: packets can take **1, 3 or 5 slots**. The radio **stays on the same frequency** for the whole packet, then returns to the frequency its hopping sequence requires for the next slot, so **2 or 4 hop frequencies are skipped**.
- **TDD prevents crosstalk** between transmit and receive in the transceiver, which is essential for a **one-chip implementation**.
- The **FH sequence is set by the master** and is a function of the **master's Bluetooth address** (permutations and XOR).
- Different piconets have different masters, hence **different hop sequences**. Occasionally two piconets use the **same physical channel in the same slot → collision and lost data**. This is rare and is handled with **FEC and error detection/ARQ**. This is **FH-CDMA** between piconets.
- **Adaptive frequency hopping**: quality metrics let the hop sequence **avoid poor channels** (e.g. those used by WiFi). Fewer than 79 channels may be used, but **at least 20**.

How can FH and TDD be combined? (BTX 14): each 625 µs slot is on a new hop frequency. The master transmits in even-numbered slots and the slave replies in the next (odd) slot **on the next hop frequency**. So **direction alternates slot by slot (TDD) while the frequency changes slot by slot (FH)**.

Microwave ovens (BTX 16): yes, they can interfere. Ovens radiate around **2.45 GHz**, inside Bluetooth's **2.4 GHz ISM** band [WCP: microwave ovens operate at 2.4 GHz]. The effect is limited because Bluetooth **hops 1600 times/s across 79 channels**, so only hops landing near the oven's frequency are hit, and **adaptive frequency hopping** can avoid them. Lost packets are recovered by **FEC/ARQ**.

MAC protocol in the baseband (BTX 17): **TDMA with master polling (FH-TDD-TDMA)**. Slaves send only in the slot after the master addresses them. **No contention within a piconet.**

10.9 ★ User logical links [BT-B, BT-S 31–35]

Link	Description
SCO (synchronous connection-oriented)	**Fixed bandwidth**, point-to-point, master ↔ **one** slave. **Reserved slots at regular intervals**; the unit is **two consecutive slots** (one each way), so it is **symmetric**. A master supports **up to 3 SCO links**; a slave supports 3 to one master or 2 to different masters. **SCO packets are never retransmitted.**
eSCO (extended SCO)	Reserves slots like SCO, but **can be asymmetric** and **supports retransmission**
ACL (asynchronous connectionless)	**Point-to-multipoint** between the master and all slaves. Uses slots **not reserved for SCO/eSCO**; the master can exchange packets with any slave **per slot**. **Only one ACL link** exists. Most ACL packets are **retransmitted** if needed

- **SCO/eSCO** carry **time-bounded data** that need a **guaranteed data rate but not guaranteed delivery**, e.g. **digitally encoded audio** tolerant of loss. The rate is guaranteed by **slot reservation**.
- **ACL** gives a **packet-switched** style connection: **no bandwidth reservation**; delivery may be guaranteed via error detection + retransmission. **A slave may send an ACL packet in the slave-to-master slot only if it was addressed in the preceding master-to-slave slot** (polling). 1-, 3- and 5-slot packets; unprotected or **2/3 FEC**. Maximum: a **5-slot unprotected** packet with asymmetric allocation gives **721 kbps forward and 57.6 kbps reverse** (Basic Rate). EDR **3-DH5** reaches **2178.1 kbps** asymmetric (Table 12.4).

10.10 Packet format [BT-B, BT-S 36–44]

```
|<--- 72 bits --->|<- 54 bits ->|<----- 0 to 2745 bits ----->|
|  Access code    |   Header    |          Payload           |

Access code: | preamble 4 | SYNC word 64 | trailer 4 |
Header (18 bits, sent with 1/3 FEC = 54):

| AM_ADDR 3 | Type 4 | Flow 1 | ARQN 1 | SEQN 1 | HEC 8 |
```

- **Access code**: timing synchronisation, offset compensation, paging, inquiry. Types: **CAC** (channel access code: identifies a piconet), **DAC** (device access code: paging and responses), **IAC** (inquiry access code: inquiry).
- **Header fields**:
  - **AM_ADDR (3 bits)**: active-mode address of a slave (at most 7 active slaves); **0 = broadcast** from the master.
  - **Type (4 bits)**: packet type (SCO/ACL types, error protection, number of slots). The **ID packet** (access code only, 68 bits) is used in inquiry and access procedures.
  - **Flow (1 bit)**: ACL flow control. 0 = stop sending, 1 = resume.
  - **ARQN (1 bit)**: acknowledgement for CRC-protected ACL traffic. **ACK = 1, NAK = 0**; no reply means a NAK is assumed.
  - **SEQN (1 bit)**: alternating 1/0 sequence bit to **filter out retransmissions** (duplicates caused by a failed ACK).
  - **HEC (8 bits)**: header error control, an **8-bit error-detection code** protecting the header.
- **Payload**: payload header (8 bits single-slot, 16 bits multislot), with **L_CH** (logical channel: LMP 11, start/unfragmented L2CAP 10, continuation 01, other 00), **Flow** (L2CAP-level flow control) and **Length** (bytes of data); the payload body; and a **16-bit CRC** on all data payloads (except AUX1).

10.11 ★ Error correction in Bluetooth [BT-S 45–50]

Three schemes:

1. **1/3-rate FEC**: used on the **18-bit header** and the voice field in **HV1** packets. It simply **sends three copies of each bit**, and **majority logic** decodes each triple.
2. **2/3-rate FEC**: used in all **DM** packets, the data field of **DV**, the **FHS** packet and **HV2**. It is a form of **Hamming code (15, 10)**: it **corrects all single errors and detects all double errors** in each codeword.
3. **ARQ**: used with **DM and DH** packets and the data field of DV. Its elements:
   - **Error detection** (CRC, supplemented by FEC)
   - **Positive acknowledgement** of error-free packets
   - **Retransmission after timeout**

- **Negative acknowledgement and retransmission** ACK/NAK are carried **in the next return packet** (piggybacked in ARQN).

10.12 ★ Master vs base station/AP, and one device as master of two piconets

Bluetooth master vs 802.11 base station/AP (Asked in: Exam Q4(a), T2-2026 Q8, BTX 8)

From [BT-S 74] (802.15.3 slide comparing a PNC, an 802.11 AP and a Bluetooth master): "*The AP provides a link to other networks and acts as a relay point for all MAC frames. The Bluetooth master sets up piconets and acts as a relay point for all traffic.*" Combined with [BT-B] and [WL]:

	Bluetooth master (central)	802.11 AP / base station
Medium access coordination	**Fully controls access**: sets the hop sequence and timing; **polls** slaves (TDD/TDMA); **a slave transmits only when addressed by the master**, so **no contention or collisions within the piconet**	Stations normally **contend** using **CSMA/CA (DCF)**: random backoff, collisions possible. Polling (**PCF**) is **optional**
Role in connecting to other networks	Sets up the **piconet** (an ad hoc PAN) and relays traffic **inside** it; it is **not a gateway** to a wired network or the Internet (slaves talk only to the master; inter-piconet communication is via scatternet bridge devices)	**Bridges the wireless BSS to the distribution system/wired LAN/Internet**; relays all MAC frames; ESS roaming
Number of devices	**Max 7 active slaves** (3-bit AM_ADDR)	Many stations (no such small fixed limit)
Nature	Any Bluetooth device can become master (role is temporary/dynamic)	Dedicated, usually fixed infrastructure device
Range / power	~10 m, low power	Tens to hundreds of metres

Can one device be the master of two piconets at the same time? (Asked in: Exam Q4(d), T2-2025 Q9, BTX 9)

No. A piconet **is defined by its master**: the hopping sequence (channel) and the timing (phase) are derived from the **master's own Bluetooth address and clock** [BT-B]. If one device were master of two "piconets", **both would use the same hopping sequence and timing**, so they would be **one and the same piconet**, not two. (Its slaves would simply all be in one piconet, limited to 7.) A device **can** be in two piconets as **slave in both**, or **master in one and slave in the other**, by **time-sharing** between the two hop sequences, which is how scatternets work [BT-B Fig 12.2].

10.13 ★ Hidden terminal problem in Bluetooth (Exam Q4(b), T2-2025 Q4)

Within a piconet, no. The hidden terminal problem arises with **contention-based** access like CSMA/CA, where stations decide on their own to transmit after sensing an idle channel. In Bluetooth the **master polls**: **a slave transmits only in the slot after the master has addressed it** [BT-B ACL rule], and SCO slots are **reserved**. Two slaves never transmit simultaneously by their own decision, so there is **nothing for a hidden terminal to collide with**. All slaves are within range of the master, which schedules every transmission (TDD/TDMA). **Between piconets** (scatternet), collisions **can** occur when two piconets happen to hop to the same frequency in the same slot. But this is **FH-CDMA collision**, not a hidden terminal problem caused by carrier sensing, and it is handled by FEC/ARQ [BT-B].

SUPPLEMENTARY (Kurose & Ross's answer to the same question): "No, the hidden terminal problem does not occur in Bluetooth because the master polls; slaves only transmit when polled."

10.14 ★ Scatternet collision probability (Asked in: Exam Q4(e), T2-2025 Q8, BTX 20)

*Two piconets, no CSMA, 80 frequencies, both always have packets to send. A successful transmission needs a successful poll (master → slave) **and** a successful reply (slave → master).*

Reasoning (uses [BT-B]: each piconet hops pseudorandomly and independently over the frequencies; a collision occurs when both use the same frequency in the same slot):

1. In any one slot, both piconets transmit (always busy). Piconet 2 is on a frequency that is effectively random relative to piconet 1. **P(same frequency in one slot) = 1/80.**
2. **P(one slot is collision-free) = 1 − 1/80 = 79/80.**
3. An exchange uses **two slots** (poll + reply) on **two different hop frequencies**, so the slots are **independent**: P(both succeed) = $(79/80)^2$ = **0.9752**.
4. **P(collision affecting the exchange) = $1 - (79/80)^2 = 159/6400 \approx 0.0248 \approx 2.5\%$.**

Answer: about 2.5% (≈ 1/80 per individual packet; ≈ 2/80 per poll-reply transaction).

SUPPLEMENTARY note on assumptions: if the question means "probability that a *single packet* collides", the answer is **1/80 = 1.25%**. State both, clearly: "per packet 1/80; per successful poll+response transaction $1 - (79/80)^2 \approx 2.5\%$". Mention that with **79** channels (real Bluetooth) the figures are 1/79 and ≈ 2.5%.

10.15 Bluetooth High Speed (3.0) and Bluetooth Smart (4.0) [BT-S 52–55]

Bluetooth 3.0 + HS: up to **24 Mbps**. It adds a **Wi-Fi connection** for higher rates: the **Alternative MAC/PHY (AMP)** (actually an 802.11 link). The Bluetooth radio still handles **device discovery, association, initial connection setup, profile configuration and connection maintenance**; AMP is used **only when higher rates are needed**. Each AMP uses a **Protocol Adaptation Layer (PAL)** to map Bluetooth protocols onto the 802.11 MAC/PHY. **(BTX 3)**: Bluetooth 3.0 reaches 24 Mbps by **using a co-located 802.11 (Wi-Fi) radio via AMP** for the data transfer.

Bluetooth Smart = Bluetooth Low Energy (BLE), Bluetooth 4.0 (BTX 2):

- An **intelligent, power-friendly** version; runs for long periods on a **single battery** or scavenged energy; can talk to legacy Bluetooth devices and smartphones; very successful for **IoT** (e.g. health monitors).
- Same **2.4 GHz** ISM band, but **40 channels spaced 2 MHz apart** (**3 advertising + 37 data**), instead of 79 × 1 MHz.
- Devices may implement transmitter, receiver or both; **single-mode** (BLE only, **cheap**, compact) or **dual-mode** (also BR/EDR).
- **Design features**: **10 mW** output; **150 m** range in an open field; new **advertising** mechanism for efficient discovery; new **asynchronous connectionless MAC** for low delay, fast transactions; new **generic attribute protocol** (simple client/server model); new **generic attribute profile** to collect sensor data efficiently.

10.16 IEEE 802.15 [BT-B 12.5, BT-S 56–80]

- The IEEE working group for **WPANs**, formed to develop standards for **short-range wireless PANs**. In a PAN all devices are **owned by one person or family** (PCs, smartphones, tablets, health monitors, smart glasses, peripherals).

- **802.15.1**: MAC and PHY from the Bluetooth specification (approved 2002, updated 2005).
- **802.15.3**: devices that are **low cost and low power** compared with 802.11 but with **higher data rates** than 802.15.1 (issued 2003). **802.15.3c**: **gigabit rates at 60 GHz**. **802.15.3d**: **100 Gbps**.
- **802.15.4**: **very low cost, very low power**, data rates **lower** than 802.15.1 (2003).
- Architecture: 802.15.1, 802.15.3 and 802.15.4 share **LLC**, with different MAC and PHY layers.

Standard	PHY / rates
802.15.1 Bluetooth	2.4 GHz; 1, 2 or 3 Mbps; 24 Mbps HS
802.15.3c	60 GHz; 1–6 Gbps
802.15.3d	100 Gbps (the slide's table labels it 60 GHz; SUPPLEMENTARY: the published standard actually uses ~252–325 GHz)
802.15.4 / 4e	868/915 MHz, 2.4 GHz; DSSS 20, 40, 100, 250 kbps; UWB 110 kbps–27 Mbps; CSS 250 kbps, 1 Mbps

802.15.3: high data rate WPAN [BT-S 63–68]

- Examples: cameras/phones to printers or kiosks; laptop to projector; **5:1 surround sound** speakers; video from a set-top box; music to headphones; camcorder to TV; remote viewfinders.
- **Requirements**: short range (~10 m); **high throughput (> 20 Mbps)** for video and multichannel audio; low power; low cost; **QoS** (guaranteed data rate); **dynamic environment** (devices enter and leave the piconet; mobile devices < 7 km/h); simple connectivity; **privacy**.
- **MAC**: a network of **DEVs**; one DEV is the **piconet coordinator (PNC)**, which sends **beacons**. Time is divided into **superframes**: a beacon period, a **contention access period** (command/control, **CSMA/CA**) and a **channel time allocation period** (**TDMA with guaranteed time slots, GTSs**: QoS).
- **★ PNC vs 802.11 AP vs Bluetooth master** [BT-S 64]: *the AP provides a link to other networks and acts as a relay point for all MAC frames; the Bluetooth master sets up piconets and acts as a relay point for all traffic;* ***the PNC controls access to the time resources of the piconet but is not involved in the exchange of data frames between DEVs***.
- **PHY**: 2.4 GHz; five modulation formats at **11 Mbaud** give **11–55 Mbps**; **trellis-coded modulation (TCM)**.
- **802.15.3c**: 60 GHz; a common ~5 GHz-wide band worldwide; **high power, doesn't penetrate walls**; **beamforming** helps; **four 2160 MHz channels**. Usage models: (1) uncompressed HD video (1920×1080 × 24 bit × 60 fps → **> 3.5 Gbps**); (2) multivideo streaming; (3) office desktop peripherals; (4) conference ad hoc; (5) kiosk file downloading. **Three PHY modes**: single carrier 1.5/3 Gbps; high-speed OFDM up to 5.775 Gbps; audio/visual OFDM 0.952/1.904/3.807 Gbps. The MAC supports coexistence of the modes, **frame aggregation**, and optional **beamforming in three stages** (sector-level coarse search, beam-level fine search, optional tracking).
- **802.15.3d**: **100 Gbps**, low cost and power; wireless offices, data-centre point-to-point links, wireless backhaul, intra-device; a few cm to 100 m.

802.15.4: low rate WPAN [BT-S 69–78]

- **TG4**: a low data rate solution with **multi-month to multi-year battery life** and **very low complexity**.
- PHYs: **868/915 MHz DSSS** (20, 40, 100, 250 kbps); **2.4 GHz DSSS** (250 kbps); **UWB** impulse radio (very short pulses, low spectral density below thermal noise; 851 kbps, optionally 110 kbps, 6.81 or 27.234 Mbps; **precision ranging to within 1 m**); **2.4 GHz chirp spread spectrum (CSS)** (1 Mbps, optionally 250 kbps).

- Applications: **sensors, interactive toys, smart badges, remote controls, home automation**.
- **MAC**: **CSMA-CA** channel access, automatic network establishment from a **coordinator node**, and a **fully handshaked** protocol for reliable delivery.
- Extensions: **4c/4d** (780 and 950 MHz PHYs); **4e** (industrial: time synchronisation, **channel hopping**, low duty cycles); **4f** (**active RFID**); **4g** (**smart utility networks/smart grid**, 40–1000 kbps); **4j** (**medical body area networks**, 2.4 GHz); **4k** (**LECIM**: critical infrastructure such as roads and bridges, multi-year battery); **4m** (**TV white space**); **4p** (**rail**: Positive Train Control).
- **Other 802.15**: **802.15.2** (**coexistence** of 802.15 with 802.11 in the same bands); **802.15.6** (**body area networks**: in-, on- and around-body; narrowband, UWB, body channel); **802.15.7** (**visible light communication**, 380–780 nm, 11.67 kbps–96 Mbps; challenges: flicker and dimming).

10.17 ZigBee [BT-B 12.6, BT-S 81–90]

- A suite of protocols **based on IEEE 802.15.4** for **small, low-power radios**, simpler than WPANs. Suited to **low data rate, long battery life, secure networking**. Data rates **20–250 kbps**; best for **periodic or intermittent** communication. Named after the **waggle dance of honey bees**. Standards since **2003** (the Alliance formed in 1998; ZigBee ratified Dec 2004; **ZigBee PRO** 2007).
- **Features**: low cost; low power; **mesh networking** (nodes route each other's messages: reliability and range); **star, tree and mesh** topologies; **one coordinator** creates, controls and maintains the network; **wakes in 30 ms** (vs Bluetooth up to 3 s), so nodes can sleep most of the time.
- **Four components added to 802.15.4**: **network layer (routing)**; **application support sublayer (APS)**; **ZigBee device objects (ZDOs)**, the most significant addition (device roles, join requests, device discovery, security); **manufacturer-defined application objects**.
- **Stack profiles**: Profile 1 (**ZigBee**: home, light commercial); Profile 2 (**ZigBee PRO**: multicasting, higher security).
- **Application profiles**: building automation, health care, home automation, input device, Light Link, network devices, retail, remote control, Smart Energy 1.1 and 2 (IP-based), telecom services. Other specs: **ZigBee IP** (IPv6 mesh); **ZigBee RF4CE** (consumer electronics remote control).
- ★ **Device types**: **Coordinator**: the most capable; **only one per network**; starts the network, is the root of a tree, maintains information such as **security keys**. **Router**: passes data on to other devices. **End device**: just enough function to talk to a router or coordinator; **cannot relay**; sleeps a lot; least memory, **cheapest**.
- **Routing**: ZigBee automatically builds low-speed **ad hoc networks**, using routing protocols to forward packets over **multiple hops**. **AODV (Ad hoc On-demand Distance Vector)** is used: it sends **route requests** and either responds or forwards them until someone responds; nodes **store and use** discovered routes and can form clusters. *(This is the only routing protocol mentioned anywhere in your notes; see Topic 11.)*
- **Networks**: **non-beacon** (unslotted CSMA/CA, routers always on, so they need permanent power) vs **beacon-based** (periodic beacons; superframe with a **contention access period** of 16 slots using CSMA/CA and an optional **contention-free period** with **GTSs**; nodes sleep between beacons, intervals 15.36 ms to 251.65824 s at 250 kbps). Beacon mode is typical for star/tree networks, non-beacon for mesh; most commercial systems use non-beacon.
- **Radio hardware**: **2.4 GHz** (16 channels, 5 MHz each, 250 kbps, **offset QPSK**); **915 MHz** (40 kbps, BPSK); **868 MHz** (20 kbps, BPSK). Range **10–75 m**, up to **1500 m with ZigBee PRO**.
- **Security**: strict key management; one **trust center**; the layer that creates a frame secures it; **128-bit keys** (network or link).

10.18 Topic 10 revision summary

- Bluetooth: Ericsson 1994; SIG; 2.4 GHz; 10 m (100 m); robust, low power, low cost; **core + profiles**. Stack: Radio, Baseband, LMP, L2CAP, SDP; RFCOMM; TCS BIN; PPP/TCP/IP/OBEX.
- **Profiles = interoperability + reuse + only-what's-needed** (asked 4 times).
- **Piconet**: 1 master + ≤ 7 active slaves; the master's address sets the hop sequence and timing; **slaves speak only when polled**. **Scatternet**: overlapping piconets; **FH-CDMA** between piconets; collision when the same hop in the same slot.
- Radio: 79 × 1 MHz, **1600 hops/s, 625 µs slots**, **GFSK 1 Mbps**, EDR 2/3 Mbps; classes 1/2/3.
- Baseband: **FH-TDD-TDMA**; 1/3/5-slot packets; adaptive FH (≥ 20 channels).
- Links: **SCO** (reserved, symmetric, no retransmission), **eSCO**, **ACL** (packet-switched, polled, retransmission).
- Packet: access code 72 / header 54 (18 × 3) / payload 0–2745; AM_ADDR, Type, Flow, ARQN, SEQN, HEC.
- Error control: **1/3 FEC (repetition), 2/3 FEC (Hamming 15,10), ARQ**.
- **Master vs AP**: polling (no contention) vs CSMA/CA; a piconet PAN vs a bridge to the wired network. **One master for two piconets is impossible**: the master defines the hop sequence. **No hidden terminal inside a piconet**: polling. **Scatternet collision ≈ $1 - (79/80)^2 \approx 2.5\%$.**
- BT 3.0 HS (AMP/802.11, 24 Mbps); BT 4.0 Smart/BLE (40 × 2 MHz channels, 10 mW, 150 m).
- 802.15.3 (high rate, PNC, superframes, GTS), .3c (60 GHz Gbps), .3d (100 Gbps), .4 (low rate, years of battery), ZigBee (coordinator/router/end device, AODV, mesh).

TOPIC 11 — ROUTING AND AD HOC NETWORK ROUTING

SUPPLEMENTARY TOPIC. None of the uploaded files cover this topic. The course outline names *Ch. 3 Communication Networks, Ch. 4 Protocols and the TCP/IP Suite* and *Ch. 13 (Agrawal) Ad hoc Networks*, which were not provided. Everything below follows the standard textbook treatment used by these exact questions: the Figure 1 network and the count-to-infinity questions are from **Kurose & Ross, *Computer Networking*, Ch. 5**; the T2-2026 Figure is **Tanenbaum's Dijkstra example**; MANET categories follow **Agrawal & Zeng** and standard MANET literature. The only related item in your notes is **AODV**, mentioned on the ZigBee slide [BT-S 87].

This topic is worth **30% of the exam paper** (and 25–32% of the Test 2 papers). Learn the Dijkstra tables below perfectly.

11.1 Routing, forwarding and switching ★ (Asked in: T2-2026 Q1, 6 marks)

Process	Purpose	Where / scope	Information used to decide
Routing	Determine the **end-to-end path** (route) packets take from source to destination	**Network-wide**, **control plane**; runs over seconds/minutes; routing algorithms (Dijkstra/link-state, Bellman-Ford/distance-vector; AODV in MANETs)	**Network topology and link costs** (exchanged with other routers via routing protocols); output = the routing/forwarding table
Packet forwarding	**Move an arriving packet from a router's input link to the correct output link**	**Router-local**, **data plane**; per packet, nanoseconds	The packet's **destination network-layer (IP) address**, looked up in the **forwarding table** (longest prefix match)
Switching	Move **frames** to the correct outgoing port **within a LAN** (link layer), or the transfer through a router's internal **switching fabric**	**Link layer** (layer 2) within one network; or inside the router	The frame's **destination MAC address**, looked up in the **switch table** (learned by **self-learning** from source addresses)

One-line distinction: routing = **planning the trip** (map → route); forwarding = **taking one interchange exit** (look up the destination in the table); switching = the **same local job at layer 2** using MAC addresses.

11.2 Graph model of routing

A network is a **graph G = (N, E)**: N = routers (nodes), E = links (edges) with **costs c(x, y)** (delay, distance, cost, or inverse bandwidth). **c(x, y) = ∞** if x and y are not neighbours. A **least-cost path** minimises the sum of link costs.

Two classes of algorithms:

- **Global (link-state)**: every node knows the **complete topology and all link costs** → runs **Dijkstra**.
- **Decentralised (distance-vector)**: each node knows **only its neighbours and their distance estimates**; computes iteratively → **Bellman-Ford**.

11.3 ★ Link-state routing and Dijkstra's algorithm

Link state: each router **floods** link-state advertisements (its neighbours and link costs) to **all** routers, so all have an identical map. Each router then runs **Dijkstra** locally (OSPF, IS-IS).

Dijkstra's algorithm (for source u):

- **D(v)**: current least-cost estimate from u to v. **p(v)**: predecessor of v on that path. **N'**: set of nodes whose least-cost path is **definitively known**.

```
Initialisation:
  N' = {u}
  for all nodes v:  if v is a neighbour of u: D(v) = c(u,v)  else D(v) = ∞
Loop (until N' = all nodes):
  find w not in N' with the minimum D(w)
  add w to N'
  for every neighbour v of w not in N':
      D(v) = min( D(v),  D(w) + c(w,v) )      ← update if the path via w is cheaper
```

- **Greedy**: always fixes the closest unfinished node. **n iterations**; complexity **O(n^2)** (O(n log n + E log n) with a heap). It needs **non-negative costs**.

Exam method: draw a table with one row per step: **Step | N' | D(v),p(v) for each node**. Mark a node's column finished once it enters N'. Finish with the **forwarding table** (least-cost path and first hop).

Figure 1 (Exam Q3(c), T2-2025 Q2) link costs

x–z 8, x–y 6, x–v 3, x–w 6, y–z 12, y–v 8, y–t 7, v–t 4, v–u 3, v–w 4, t–u 2, w–u 3.

Adjacency (read this off Figure 1 before you start, and write it in your answer):

Node	Neighbours (cost)
x	z 8, y 6, v 3, w 6
y	z 12, x 6, v 8, t 7
z	x 8, y 12
v	x 3, y 8, w 4, u 3, t 4
w	x 6, v 4, u 3
u	v 3, w 3, t 2
t	y 7, v 4, u 2

★ Worked example DJ1: shortest paths from x (T2-2025 Q2, 8 marks)

Step	N'	D(t),p	D(u),p	D(v),p	D(w),p	D(y),p	D(z),p
0	x	∞	∞	**3,x**	6,x	6,x	8,x
1	xv	7,v	**6,v**	—	6,x	6,x	8,x
2	xvu	7,v	—	—	**6,x**	6,x	8,x
3	xvuw	7,v	—	—	—	**6,x**	8,x
4	xvuwy	**7,v**	—	—	—	—	8,x
5	xvuwyt	—	—	—	—	—	**8,x**
6	xvuwytz	—	—	—	—	—	—

Step-by-step reasoning:

- Step 0: x's neighbours v 3, w 6, y 6, z 8; t and u are ∞. Minimum = **v (3)**.
- Step 1 (via v, D = 3): t = 3 + 4 = **7**; u = 3 + 3 = **6**; w: 3 + 4 = 7 > 6, keep 6; y: 3 + 8 = 11 > 6, keep. Minimum among u 6, w 6, y 6 → pick **u** (tie broken alphabetically).
- Step 2 (via u, D = 6): t: 6 + 2 = 8 > 7, keep 7; w: 6 + 3 = 9 > 6, keep. Add **w (6)**.
- Step 3 (via w): no improvements (u, v done). Add **y (6)**.
- Step 4 (via y): t: 6 + 7 = 13 > 7; z: 6 + 12 = 18 > 8. Add **t (7)**.
- Step 5: add **z (8)**.

Result (least-cost paths from x):

Destination	Cost	Path	First hop (forwarding table)
v	3	x → v	v
u	6	x → v → u	v
w	6	x → w	w
y	6	x → y	y
t	7	x → v → t	v
z	8	x → z	z

★ Worked example DJ2: shortest paths from t (Exam Q3(c)(i), 4 marks)

Step	N'	D(u),p	D(v),p	D(w),p	D(x),p	D(y),p	D(z),p
0	t	**2,t**	4,t	∞	∞	7,t	∞
1	tu	—	**4,t**	5,u	∞	7,t	∞
2	tuv	—	—	**5,u**	7,v	7,t	∞
3	tuvw	—	—	—	**7,v**	7,t	∞
4	tuvwx	—	—	—	—	**7,t**	15,x
5	tuvwxy	—	—	—	—	—	**15,x**
6	tuvwxyz	—	—	—	—	—	—

Key updates: via u (2): w = 2 + 3 = 5; v: 2 + 3 = 5 > 4, keep. Via v (4): x = 4 + 3 = **7**; w: 4 + 4 = 8 > 5; y: 4 + 8 = 12 > 7. Via w (5): x: 5 + 6 = 11 > 7. Via x (7): z = 7 + 8 = **15**; y: 7 + 6 = 13 > 7. Via y (7): z: 7 + 12 = 19 > 15.

Destination	Cost	Path
u	2	t → u
v	4	t → v
w	5	t → u → w
x	7	t → v → x
y	7	t → y
z	15	t → v → x → z

★ Worked example DJ3: shortest paths from w (Exam Q3(c)(ii), 4 marks)

Step	N'	D(t),p	D(u),p	D(v),p	D(x),p	D(y),p	D(z),p
0	w	∞	**3,w**	4,w	6,w	∞	∞
1	wu	5,u	—	**4,w**	6,w	∞	∞
2	wuv	**5,u**	—	—	6,w	12,v	∞
3	wuvt	—	—	—	**6,w**	12,v	∞
4	wuvtx	—	—	—	—	**12,v**	14,x
5	wuvtxy	—	—	—	—	—	**14,x**
6	wuvtxyz	—	—	—	—	—	—

Key updates: via u (3): t = 3 + 2 = 5; v: 3 + 3 = 6 > 4. Via v (4): y = 4 + 8 = 12; t: 4 + 4 = 8 > 5; x: 4 + 3 = 7 > 6. Via t (5): y: 5 + 7 = 12 (equal, keep v). Via x (6): z = 6 + 8 = 14; y: 6 + 6 = 12 (equal). Via y (12): z: 12 + 12 = 24 > 14.

Destination	Cost	Path
u	3	w → u
v	4	w → v
t	5	w → u → t
x	6	w → x
y	12	w → v → y **(tie: w → x → y and w → u → t → y also cost 12)**
z	14	w → x → z

> **Tie note**: y has **three equal-cost paths (12)**. Any one is correct; state the tie-breaking rule you used (e.g. "keep the first-found predecessor"). Mentioning the tie shows full understanding.

★ Worked example DJ4: T2-2026 Q9 (A to all nodes, 10 marks)

Link costs read from the figure: **A–B 2, A–G 6, B–C 7, B–E 2, E–G 1, E–F 2, F–C 3, F–H 2, G–H 4, C–D 3, H–D 2.** (This is Tanenbaum's classic Dijkstra example.)

Node	Neighbours (cost)
A	B 2, G 6
B	A 2, C 7, E 2
C	B 7, F 3, D 3
D	C 3, H 2
E	B 2, F 2, G 1
F	E 2, C 3, H 2
G	A 6, E 1, H 4
H	G 4, F 2, D 2

Step	N'	B	C	D	E	F	G	H
0	A	**2,A**	∞	∞	∞	∞	6,A	∞
1	AB	—	9,B	∞	**4,B**	∞	6,A	∞
2	ABE	—	9,B	∞	—	6,E	**5,E**	∞
3	ABEG	—	9,B	∞	—	**6,E**	—	9,G
4	ABEGF	—	9,B	∞	—	—	—	**8,F**
5	ABEGFH	—	**9,B**	10,H	—	—	—	—
6	ABEGFHC	—	—	**10,H**	—	—	—	—
7	ABEGFHCD	—	—	—	—	—	—	—

Step-by-step reasoning:

- Step 0: neighbours of A: B 2, G 6. Pick **B (2)**.
- Step 1 (via B = 2): C = 2 + 7 = 9; E = 2 + 2 = **4**. Pick **E (4)**.
- Step 2 (via E = 4): G = 4 + 1 = **5** < 6 → update G to (5, E); F = 4 + 2 = 6. Pick **G (5)**.
- Step 3 (via G = 5): H = 5 + 4 = 9. Pick **F (6)**.
- Step 4 (via F = 6): C: 6 + 3 = 9 = 9 (**tie**, keep B); H = 6 + 2 = **8** < 9 → (8, F). Pick **H (8)**.
- Step 5 (via H = 8): D = 8 + 2 = 10. Pick **C (9)**.
- Step 6 (via C = 9): D: 9 + 3 = 12 > 10, keep. Pick **D (10)**.

(b) Final costs and routes from A:

Destination	Cost	Complete shortest route
B	**2**	A → B
C	**9**	A → B → C *(equal-cost alternative: A → B → E → F → C, also 9)*
D	**10**	A → B → E → F → H → D
E	**4**	A → B → E
F	**6**	A → B → E → F
G	5	A → B → E → G (note: cheaper than the direct link A–G = 6)
H	8	A → B → E → F → H

11.4 ★ Distance-vector routing and Bellman-Ford

Bellman-Ford equation: the least cost from x to y is

$$d_x(y) = \min \text{ over all neighbours v of } \{\ c(x,v) + d_v(y)\ \}$$

Distance-vector algorithm (RIP): each node keeps a **distance vector D_x = [$D_x(y)$ for all y]** and its neighbours' vectors.

1. **Initialise**: $D_x(y)$ = c(x, y) for neighbours, ∞ otherwise; send D_x to the neighbours.
2. **Loop**: wait for a **link-cost change** or a **neighbour's new vector**; recompute $D_x(y) = \min_v\{c(x, v) + D_v(y)\}$; **if D_x changed, send it to the neighbours**.

It is **iterative** (until no more changes), **asynchronous** (no lock-step) and **distributed** (each node uses only neighbour information).

★ Link-state vs distance-vector (Exam Q3(a), 4 marks)

	Link-state (LS)	Distance-vector (DV)
Knowledge	**Global**: every router has the **complete topology**	**Local**: only neighbours' cost estimates ("routing by rumour")
Algorithm	**Dijkstra**	**Bellman-Ford**
What is sent, to whom	Own **link costs**, **flooded to all** routers	Whole **distance vector**, **only to neighbours**
Message complexity	High: O(n·E) messages per change	Lower: exchanges only between neighbours
Speed of convergence	Fast: $O(n^2)$ computation, **no loops** once floods finish	**Slow**, variable; **routing loops** possible; **count-to-infinity**
Robustness	Better: each router computes its **own** table; a bad advertisement affects only its own link	Worse: a wrong cost **propagates through the whole network**
Examples	OSPF, IS-IS	RIP, (BGP is path-vector)

Similarities ("compare"): both find **least-cost paths**, both produce a **forwarding table**, both are **dynamic** (adapt to cost changes) and both use **link costs**.

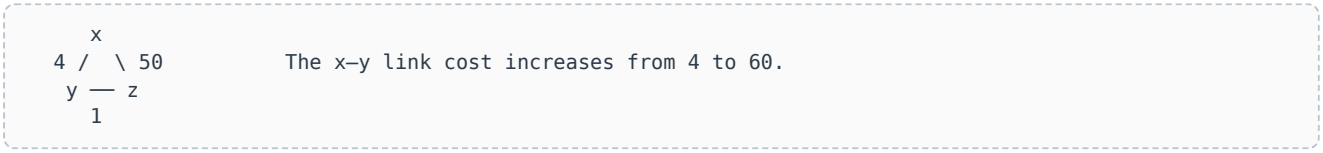
★ Essential difference: Dijkstra vs Bellman-Ford (Exam Q3(b), 4 marks)

- **Dijkstra** is a **centralised/global, greedy** algorithm: the node computing it needs the **entire graph**. It **permanently fixes one node per iteration** (the nearest unfinished node), so it finishes in **n steps**. It requires **non-negative** link costs.
- **Bellman-Ford** is **decentralised and iterative**: each node computes using **only its neighbours' current distance estimates** and **repeatedly relaxes** distances until nothing changes (at most n − 1 rounds). It **handles negative link costs** (and can detect negative cycles), is naturally **distributed**, but **converges more slowly** and can suffer count-to-infinity.
- **In one sentence**: *Dijkstra uses complete global knowledge and settles nodes greedily one at a time; Bellman-Ford uses only neighbour information and iteratively relaxes all estimates until they converge.*

11.5 ★ The count-to-infinity problem (Exam Q3(d), T2-2025 Q3)

What it is: in distance-vector routing, **bad news (a link-cost increase or failure) travels slowly**. Neighbours keep advertising **stale** routes that actually pass through the node that has just lost the route, which creates a **routing loop**. The nodes keep increasing their estimates step by step, "counting" up towards the true cost, or **towards infinity** if the destination is unreachable.

Classic example (Kurose & Ross): y–x cost 4, y–z 1, z–x 50. Initially D_y(x) = 4, D_z(x) = 5 (via y).



1. y sees c(y, x) = 60. y computes D_y(x) = min{60 + 0, c(y,z) + D_z(x)} = min{60, 1 + 5} = **6**. That is wrong: z's 5 is a route **through y**! Now y routes to x **via z**, and z routes to x **via y**: a **routing loop**.
2. y tells z; z updates D_z(x) = min{50, 1 + 6} = **7**.
3. z tells y; y updates to 8; z to 9; ...
4. This continues for **44 iterations**, until z's estimate via y exceeds 50 and z finally chooses its direct link (50); then y = 51. Packets **loop between y and z** the whole time. If the x–y link had **failed** (cost ∞) and z had no other route, they would **count to infinity**.

Fixes: **poisoned reverse** (if z routes to x through y, z tells y D_z(x) = ∞; this fixes loops between **two** nodes but **not** larger loops); **split horizon**; defining **infinity as a small number** (RIP: 16 hops); hold-down timers.

★ **Will it occur if we decrease the cost of a link? No.** Decreasing a cost is **good news**, and **good news travels fast**. Each node that hears of the lower cost computes a lower (correct) distance through it and sends it on. Distances only **decrease** and converge within a number of iterations bounded by the network diameter. No node relies on a stale path that is now too cheap, so no loop forms. (Kurose: *"No. Decreasing link cost won't cause a loop."*)

★ **What about connecting two nodes that have no link? No.** Adding a new link is the same as **decreasing its cost from ∞ to a finite value**: it is good news, which propagates quickly without counting to infinity.

11.6 ★ Ad hoc networks (MANETs) and why routing is complicated (Exam Q2(d))

A **MANET** (mobile ad hoc network) is a collection of **mobile nodes with no fixed infrastructure** (no AP or base station). Nodes communicate directly if in range, otherwise **other nodes forward** packets (multi-hop). **Every node is also a router.** This matches your WLAN notes: in ad hoc networks *"nodes can only communicate if they can reach each other physically... or if other nodes can forward the message"*, and *"the complexity of each node is higher: implement medium access mechanisms, forwarding data"* [WL 8–9].

Why routing in ad hoc networks is complicated:

1. **Dynamic topology**: nodes **move**, join and leave, so links **break and form constantly**; routes go stale quickly.
2. **No infrastructure or central control**: no dedicated routers; **every node must route**, with **distributed** decisions.
3. **Limited bandwidth**: the shared wireless channel is scarce, so routing control traffic (flooding updates) competes with data.
4. **Limited battery and processing** in mobile nodes, so frequent route computation and transmission drain energy.
5. **Unreliable wireless links**: fading, interference, **hidden/exposed terminals**, high error rates, sometimes **asymmetric (unidirectional)** links.
6. **Multi-hop**: many hops, so delays and failure points multiply.
7. **Scalability and security**: large networks generate huge overhead; no trusted infrastructure, so it is vulnerable to misbehaving nodes.

11.7 ★ The three categories of MANET routing protocols (Exam Q2(d) 10, T2-2025 Q1 12)

	1. Proactive (table-driven)	2. Reactive (on-demand / source-initiated)	3. Hybrid
Principle	Every node **continuously maintains routes to all destinations** in routing tables, updated by **periodic and triggered** broadcasts, whether or not routes are needed	Routes are discovered **only when a source needs one**: **route discovery** (flood a route request, the destination or an intermediate node replies) + **route maintenance** (repair broken routes)	**Combines both**: **proactive within a local zone** (nearby nodes), **reactive between zones** (distant nodes)
Route availability	**Immediately available**, so **low latency**	**Delay** while discovering the route	Fast locally; on-demand far away
Overhead	**High, constant** control traffic; wasted on unused routes; poor with high mobility	**Low** when traffic is light; but route-discovery floods are costly under heavy traffic	Balanced; tunable zone size
Best for	Small, low-mobility networks with frequent traffic	Large, highly mobile networks with occasional traffic	Large networks with mixed traffic
Examples	**DSDV** (Destination-Sequenced Distance Vector), **OLSR** (Optimized Link State Routing), WRP	**AODV** (Ad hoc On-demand Distance Vector, as used by **ZigBee** [BT-S 87]), **DSR** (Dynamic Source Routing), TORA	**ZRP** (Zone Routing Protocol)

AODV in brief (your notes [BT-S 87] + standard detail): a source broadcasts a **RREQ (route request)**; intermediate nodes record the **reverse path** and rebroadcast; the destination (or a node with a fresh route) **unicasts an RREP (route reply)** back along the reverse path, which sets up forward entries; **sequence numbers** prevent loops and stale routes; **RERR** messages report broken links.

DSDV in brief: a table-driven distance-vector protocol (Bellman-Ford). Each entry carries a **destination sequence number** to avoid loops and count-to-infinity; full and incremental updates are broadcast periodically.

SUPPLEMENTARY: some textbooks (incl. Agrawal) name the categories **table-driven, source-initiated on-demand, and hybrid**. These are the same three. Others add **geographic/position-based** or hierarchical routing.

11.8 Topic 11 revision summary

- **Routing** (network-wide path, topology and costs, control plane) vs **forwarding** (router-local, destination IP + forwarding table) vs **switching** (layer 2, MAC + switch table).
- **Link-state**: flood link costs, everyone has the map, **Dijkstra** (greedy, global, $O(n^2)$, non-negative costs). **Distance-vector**: exchange vectors with neighbours, **Bellman-Ford** $d_x(y) = \min_v\{c(x,v) + d_v(y)\}$, iterative, asynchronous, distributed; slow; count-to-infinity.
- **Dijkstra answers**: from **x**: v3, u6, w6, y6, t7, z8. From **t**: u2, v4, w5, x7, y7, z15. From **w**: u3, v4, t5, x6, y12 (3-way tie), z14. **T2-2026 from A**: B2, E4, G5, F6, H8, C9 (tie), D10.
- **Count to infinity**: bad news travels slowly → loops (44 iterations in the 4/1/50 example). **Cost decrease or a new link: no** (good news travels fast). Fixes: poisoned reverse, split horizon, max hop count.

- **MANET routing is hard**: mobility, no infrastructure, bandwidth, energy, unreliable links. **Three categories: proactive (DSDV, OLSR), reactive (AODV, DSR), hybrid (ZRP).**